

JILLIAN AWE

jilliana2600@gmail.com • www.linkedin.com/in/jillianawe • jilliana2600.github.io

SUMMARY

- Seeking a summer 2027 electrical engineering internship focused on schematic design, PCB design and hardware development with meaningful hands-on project opportunities
- Experience with schematic capture and PCB layout using Xpediton and Alitum, with hands-on experience in PCB prototyping, soldering, circuit testing and troubleshooting

EDUCATION

University of Wisconsin-Platteville, Platteville, WI

Anticipated Graduation: May 2028

Bachelor of Science, Electrical Engineering (ABET Accredited)

GPA 4.00/4.00

Emphases: Computer Engineering; Controls

Chancellor's List (4 semesters); Dean's List (4 semesters)

Clubs/Organizations: Pioneer Robotics, Marching Pioneers, Women in STEM Center, Society of Women Engineers

TECHNICAL SKILLS

Hardware: Soldering (through-hole, SMD), Multimeter, Oscilloscope, Power Supply, PCB Prototyping

Software: Xpediton, Altium, SolidWorks, Microsoft Office, KiCAD, LTSpice

Programming Languages: Python, C, C++, MATLAB, ARM Assembly, HTML, JavaScript

RELEVANT COURSEWORK

Automatic Controls • Computer Architecture • Differential Equations • Logic and Digital Design • Analog Electronics
Computational Methods • Modern Physics • Signals and Systems • Circuit Modeling • Introduction to Electrical Machines

PROFESSIONAL EXPERIENCE

Electrical Engineering Intern | Banner Engineering | *Plymouth, MN*

Summer, 2026

- Redesigned and rebuilt a legacy test fixture from over 20-year-old schematics, replacing obsolete components and adding new functionality requested by testing team
- Created schematic and PCB layout in Xpediton and collaborated with electrical engineers, mechanical engineers and PCB designers through design reviews and manufacturing
- Tested PCB using digital multimeter, oscilloscope and power supplies, identifying hardware issues through schematic and datasheet analysis and implementing modifications to resolve design issues
- Developed and documented engineering changes through Banner's NPD process, including ECOs, BOM management, missing-part reports, and design documentation to ensure future reference and maintenance
- Troubleshoot and repaired a second testing fixture by identifying circuit design problems, correcting schematic and PCB layout and creating and testing a working design
- Integrated both fixtures into a larger automated testing system using a DE10 Standard FPGA development board and Python to create a custom user interface for test operation

INVOLVEMENT

Pioneer Robotics (VEX U Robotics) | UW-Platteville | *PLATTI Driftless*

2024 - Present

Member: 2024-Present | VEX U Coordinator: 2025-2026 | Secretary: 2026-Present

- Designed and built competition robots as part of an interdisciplinary engineering team, competing at VEX U Worlds and earning the Excellence Award at the largest VEX U qualifying event and the Think Award at Worlds
- Researched, designed and integrated a custom optical odometry system using a SparkFun optical tracking sensor, developing interface circuitry to connect the sensor to the VEX Brain, working with the software team to integrate into the robot for improved autonomous navigation and troubleshooting grounding problems
- Designed custom PCB in KiCAD for the odometry system, adding ESD, surge and input protection
- Coordinated VEX U teams as VEX U Coordinator and Secretary, organizing team activities, competition logistics and communicating between team members while integrating new members into teams and getting them involved

- Designed a four-layer PCB for a line-following robot in Altium Designer, working on schematic design, component selection and placement, routing, power connections and grounding
- Designed the schematic and PCB layout for added navigation feature using IMU and two optical wheel encoders, including SPI connections and signal conditioning for the encoder outputs
- Evaluated component selection, power consumption, PCB layout and sensor integration while documenting design decisions and engineering tradeoffs
- Placed runner-up in the PCB and schematic design round out of 31 teams from universities around the country

Marching Pioneers | UW-Platteville

2024 - Present

- Committed approximately 12 hours per week during marching season to rehearsals, performances and preparation while balancing academic and extracurricular activities
- Performed at home football games and local parades, as well as at Lambeau Field and a Milwaukee Brewers game

Society of Women Engineers | UW-Platteville

2024 - Present

- Planned and budgeted STEM Expo activities for hundreds of students from area middle and high schools as a chair

WORK EXPERIENCE**Undergraduate Research Assistant** | UW-Platteville Dept. of Electrical & Computer Engineering

Summer, 2026

- Analyzed gender representation and dataset bias across computer vision datasets including FairFace, CelebA and UTKFace using Python, pandas, NumPy and Jupyter
- Developed Python-based data analysis workflows to process dataset annotations, calculate demographic distributions and model performance metrics, and visualize differences across demographic groups
- Evaluated model accuracy and error rates across demographic groups and documented findings through reproducible Jupyter notebooks and technical reports

Women in STEM Student Assistant / Peer Mentor | UW-Platteville Student Success Programs

2026 - Present

- Planned and supported events for college and pre-college students, promoting STEM education and career opportunities while creating an inclusive and supportive environment for students interested in STEM
- Served on event planning committees and helped organize STEM outreach events, campus visits and student programs for both current and prospective students and industry members
- Mentored and supported STEM students through academic and career exploration, course selection, professional development and transitioning to college, helping students connect with campus resources and explore opportunities

Student Tutor | UW-Platteville Tutoring Center

2025 - 2026

- Tutored undergraduate students in Circuit Modeling I and II, Physics II, Logic and Digital Design and math classes starting at Elementary Algebra through Differential Equations, reinforced core engineering concepts and developed clear technical communication skills
- Adapted explanations to individual learning styles, demonstrated patience, problem-solving, and learned technical material

Guest Service / Food Service Coworker | Kwik Trip | *Eau Claire, WI*

2024 - 2026

- Handled a high volume of customer interactions with professionalism and collaborated with coworkers to meet customer demands, developing strong communication, teamwork, and problem-solving skills under pressure
- Maintained equipment, including gas pumps, registers, coffee machines, and more, followed safety protocols, and adapted quickly to operational issues
- Demonstrated reliability, punctuality, and strong work ethic in a fast-paced environment

HONORS**Valedictorian** | Eleva-Strum Central High School

2024

AP Scholar with Distinction | College Board

2024